1 BPPC-IF: Bin Packing Problem with Conflicts and Item Fragmentation

1.1 Sets and parameters

- V: set of items;
- E: set of conflicting pairs of items;
- $\mathcal{U}$: set of bins;
- w_i: size of item $i \in V$;
- W: capacity of each bin.

The set of bins is defined as:

$$U = \left\lceil \sum_{i \in V} \frac{w_i}{W} \right\rceil$$

$$\mathcal{U} = \{1, 2, \ldots, U\}$$

1.2 Decision variables

$$x_{ij} = \begin{cases} 1 & \text{if a fragment of item } i \text{ is packed into bin } j, \\ 0 & \text{otherwise.} \end{cases} \qquad i \in V,\ j \in \mathcal{U}$$

$$f_{ij} = \text{size of the fragment of item } i \text{ packed into bin } j \qquad i \in V,\ j \in \mathcal{U}$$

$$y_j = \begin{cases} 1 & \text{if bin } j \text{ is used,} \\ 0 & \text{otherwise.} \end{cases} \qquad j \in \mathcal{U}$$

1.3 Mathematical formulation

$$\text{MIP:} \qquad \min \sum_{j \in \mathcal{U}} y_j \tag{1}$$

subject to:

$$\sum_{i \in V} f_{ij} \leq W y_j \qquad j \in \mathcal{U} \tag{3}$$

$$\sum_{j \in \mathcal{U}} f_{ij} = w_i \qquad i \in V \tag{4}$$

$$f_{ij} \leq w_i x_{ij} \qquad i \in V,\ j \in \mathcal{U} \tag{5}$$

$$x_{ij} + x_{kj} \leq y_j \qquad (i,k) \in E,\ j \in \mathcal{U} \tag{6}$$

$$\sum_{i \in V} f_{ij} \leq \sum_{i \in V} f_{i,j-1} \qquad j \in \mathcal{U} \setminus \{1\} \tag{7}$$

$$x_{ij} \leq y_j \qquad i \in V,\ j \in \mathcal{U} \tag{8}$$

$$x_{ij} \in \{0,1\} \qquad i \in V,\ j \in \mathcal{U} \tag{9}$$

$$f_{ij} \geq 0 \qquad i \in V,\ j \in \mathcal{U} \tag{10}$$

$$y_j \in \{0,1\} \qquad j \in \mathcal{U} \tag{11}$$